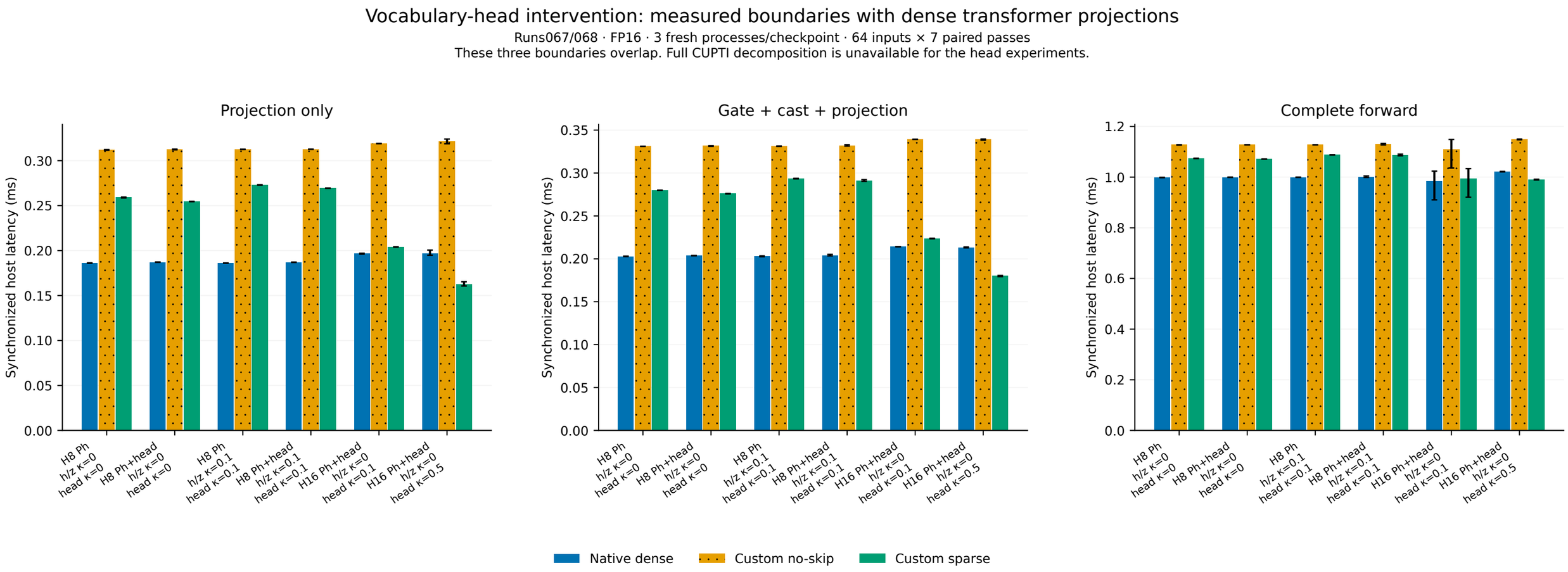
Vocabulary-head intervention: measured boundaries with dense transformer projections
Runs067/068 · FP16 · 3 fresh processes/checkpoint · 64 inputs × 7 paired passes
These three boundaries overlap. Full CUPTI decomposition is unavailable for the head experiments.
Projection only
Gate + cast + projection
Complete forward
Synchronized host latency (ms)
H8 Ph h/z κ=0
H8 Ph+head h/z κ=0 head κ=0
H8 Ph h/z κ=0.1
H8 Ph+head h/z κ=0.1 head κ=0.1
H16 Ph+head h/z κ=0 head κ=0.1
H16 Ph+head h/z κ=0 head κ=0.5
Native dense
Custom no-skip
Custom sparse